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[23ESCE11]

FIRST SEMESTER B.Tech EXAMINATION, JANUARY - 2024
FUNDAMENTALS OF CIVIL ENGINEERING AND MATERIALS

Time: 3 Hours **Maximum Marks: 100**

Instructions: i. Missing data may be suitably assumed.

ANSWER ALL QUESTIONS

		PART-A	**05 X 02=10**		
1.	a	Enumerate the different specialties within the field of Civil Engineering.	02	01	**02**
	b.	Explain the purpose of road	02	02	**02**
	c.	Highlight the essential role of cement in the construction industry.	02	03	**02**
	d.	Define and explain brick masonry	02	04	**02**
	e.	What does the term "comprehensive development plan" entail?	02	05	**02**
		PART-B	**06 X 05=30**		
2.	a.	Write short notes on: i. Geotechnical Engineering ii. Environmental Engineering	02	01	**05**
		OR			
	b.	Comment on i. Surveying ii. Water supply engineering	02	01	**05**
3.	a.	Examine the pivotal role played by civil engineers in influencing and propelling the progress of infrastructure development.	03	02	**05**
		OR			
	b.	How infrastructure influences the socio-economic advancement of a country.	03	02	**05**
4.	a.	Examine the uses and advantages of concrete as a building material.	03	03	**05**
		OR			
	b.	Provide a comprehensive explanation of Bogues compounds.	03	03	**05**
5.	a.	Differentiate between stretcher bond and header bond	04	04	**05**
		OR			
	b.	Outline the criteria for a well-constructed floor.	04	04	**05**
6.	a.	Distinguish between lintels and arches.	04	04	**05**
		OR			

	b.	What are the requirements of a good stone masonry?	04	04	**05**
7.	**a.**	Outline the different classifications of residential buildings.	04	05	**05**
		OR			
	b.	Explain the significance of green buildings.	04	05	**05**
		PART-C	**06 X 10=60**		
8.	**a.**	Outline the specific goals of the National Building Code and provide an in-depth explanation of the diverse types of buildings categorized under its framework.	02	01	**10**
		OR			
	b.	Write short notes on: **i.** Hydraulics Engineering **ii.** Construction technology **iii.** Irrigation Engineering **iv.** Engineering Geology	02	01	**10**
9.	**a.**	Provide detailed explanations for four different categorizations of roads.	02	02	**10**
		OR			
	b.	Provide detailed explanations for four different bridge types.	02	02	**10**
10.	**a.**	Elaborate on the comprehensive categorization of bricks.	03	03	**10**
		OR			
	b.	Differentiate between fine and coarse aggregates. Explain any three of their properties	03	03	**10**
11.	**a.**	Explain the different types of cement	03	03	**10**
		OR			
	b.	Define the role of paint and elucidate the diverse categories of paints.	03	03	**10**
12.	**a.**	Describe the various categories of roofing systems.	04	04	**10**
		OR			
	b.	Elaborate on the various classifications of flooring options.	04	04	**10**
13.	**a.**	Illustrate the importance of site selection in construction.	04	05	**10**
		OR			
	b.	Examine the importance of sanitation and plumbing systems in buildings.	04	05	**10**